01 · LESSON

Plant adaptations: roots, stems and leaves

9B Plant growth · Lesson 2 of 7

Date _____ Class _____

Today I can

- ☐ I can describe what plants need from the soil and the air, and how roots, root hair cells and xylem are adapted to supply water and mineral salts.
- ☐ I can explain how a leaf and its palisade cells are adapted for photosynthesis.

What happens today

1. Do Now: where do the reactants come from?
2. Learn: What plants need from mineral salts · Roots: collecting water and mineral salts · The route of water through a plant · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 24 to 25 · Workbook pages 17

What I learned today (one or two sentences)

02 · KEY VOCABULARY

Key Vocabulary

Term	Definition
mineral salts	substances dissolved in soil water that plants need to stay healthy, e.g. nitrates
nitrate	a mineral salt that supplies nitrogen for making proteins
magnesium	an element plants need to make chlorophyll
potassium	an element root cells need to absorb water properly
root hair cell	a root cell with a long extension that gives a large surface area for absorbing water
xylem	hollow tubes made from dead cells that carry water and mineral salts up the plant
wilt	to droop because the cells have too little water
palisade cell	a leaf cell near the upper surface that is packed with chloroplasts
epidermis	the outer layer of cells of a leaf
cuticle	the waxy layer on a leaf that stops it losing too much water
surface area	the total area of the outside of something

Quick Review

where do the reactants come from?

1 Write the word equation for photosynthesis.

2 Which reactant comes from the air, and which comes from the soil?

3 Name two other things a plant needs that it gets from the soil.

Core 1 of 2

Everyone does these. Answer on the card or under it.

1 Complete the table to show how plants get what they need for growth. Use the word bank: air, soil water, respiration, photosynthesis, making proteins, making chlorophyll, absorbing water, keeping cells firm.

Substance	Where it comes from	How the plant uses it
oxygen		
carbon dioxide		
nitrate salts		
magnesium salts		
potassium salts		
water		

2 Match each part of a leaf to how it helps the plant. Draw a line.

palisade cells	carries water and mineral salts from the roots
phloem	stops the leaf losing too much water
xylem	allow gases to enter and leave the leaf
stomata	contain many chloroplasts, near the top where there is most light
cuticle	carries dissolved sugars from the leaf to other parts of the plant

Core 2 of 2

Everyone does these. Answer on the card or under it.

3 Complete the sentences about the route of water. Use the word bank.

Water enters the plant through _____ cells, which have a large _____ so they absorb water quickly.

The water travels up the root and stem in _____ vessels, which are hollow tubes made from _____ cells.

If a plant has too little water, its cells sag and the plant _____.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 The Venus flytrap grows in bogs with very few mineral salts in the soil. It catches insects when the trigger hairs in its trap are touched more than once.

a) Suggest why a trigger hair has to be touched more than once before the trap shuts.

b) State what the Venus flytrap gets from the insects, and explain why it might die without them even though it can photosynthesise.

07 · CHALLENGE

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Describe how a leaf is adapted to allow its palisade cells to get all the resources they need for photosynthesis: light, carbon dioxide and water.

08 · EXIT TICKET

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Name the mineral salt a plant needs to make proteins.

2 How are root hair cells adapted to absorb water?

3 Why do palisade cells contain so many chloroplasts?

Today I can: Goal 1 ☐ Goal 2 ☐